

VENSTAR®

COMMERCIAL MODEL

DIGITAL THERMOSTAT

Up To 2 Heat & 2 Cool Stages
with Humidity Control

T4700



EXPLORER®

Simply connected.

*Anytime. Anywhere.**

Optional accessories
available, including Wi-Fi



OWNER'S MANUAL AND INSTALLATION INSTRUCTIONS





CAUTION

Follow the Installation Instructions before proceeding. Set the thermostat mode to “OFF” prior to changing settings in setup or restoring Factory Defaults.

FCC Compliance Statement

This equipment has been tested and found to comply with the limits for an intentional radiator, pursuant to Part 15, subpart C of the FCC rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference in radio communications. However, there is no guarantee that the interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that of the receiver.
- Consult the dealer or an experienced radio or TV technician for help.

Notice: Only peripherals complying with FCC limits may be attached to this equipment. Operation with noncompliant peripherals or peripherals not recommended by Venstar, is likely to result in interference to radio and TV reception. Changes or modifications to the product, not expressly approved by Venstar could void the user's authority to operate the equipment.

FCC - INDOOR Mobile Radio Information:

To comply with FCC/IC RF exposure limits for general population / uncontrolled exposure, the antenna(s) used for this transmitter must be installed to provide a separation distance of at least 20 cm from all persons and must not be co-located or operating in conjunction with any other antenna or transmitter.

This Device complies with Industry Canada License-exempt RSS standard(s). Operation is subject to the following two conditions: 1) this device may not cause interference, and 2) this device must accept any interference, including interference that may cause undesired operation of the device.

Under Industry Canada regulations, this radio transmitter may only operate using an antenna of a type and maximum (or lesser) gain approved for the transmitter by Industry Canada. To reduce potential radio interference to other users, the antenna type and its gain should be so chosen that the equivalent isotropically radiated power (e.i.r.p.) is not more than that necessary for successful communication.

Cet appareil est conforme avec Industrie Canada, exempts de licence standard RSS(s). Son fonctionnement est soumis aux deux conditions suivantes: 1) ce dispositif ne doit pas causer d'interférences, et 2) ce dispositif doit accepter toute interférence, y compris les interférences qui peuvent causer un mauvais fonctionnement de l'appareil.

En vertu des règlements d'Industrie Canada, cet émetteur de radio ne peut fonctionner en utilisant une antenne d'un type et maximale (ou moins) Gain approuvé pour l'émetteur par Industrie Canada. Pour réduire les interférences radio potentielles aux autres utilisateurs, le type d'antenne et son gain doivent être choisis afin que la puissance isotrope rayonnée équivalente (PIRE) ne est pas plus de ce qui est nécessaire pour une communication réussie.

We, Venstar, declare under our sole responsibility that the device to which this declaration relates: Complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

FCC ID: MUH-SKYPOR3

IC: 12547A-SKYPOR3

This Explorer thermostat has the ability to receive updates to its firmware. Periodically firmware updates are released by the manufacturer to add features and/or performance enhancements. This manual was produced reflecting the most current firmware/feature set at the time of publication, firmware rev. 12. Firmware releases after rev. 12 may not be adequately depicted in this manual. Please refer to the appropriate website or contact your place of purchase to learn about changes to the thermostat after firmware release 12.



Industry
Canada

Industrie
Canada

Glossary of Terms

Auto-Changeover: A mode in which the thermostat will turn on the heating or cooling based on room temperature demand.

Cool Setpoint: The warmest temperature that the space should rise to before cooling is turned on (without regard to deadband).

Deadband: The number of degrees the thermostat will wait, once a setpoint has been reached, before energizing heating or cooling.

Dehumidify: To reduce the amount of moisture in the air.

Differential: The forced temperature difference between the *heat setpoint* and the *cool setpoint*.

Heat Setpoint: The coolest temperature that the space should drop to before heating is turned on (without regard to deadband).

Humidify: To increase the amount of moisture in the air.

Icon: The word or symbol that appears on the thermostat display.

Mode: The current operating condition of the thermostat (i.e. Off, Heat, Cool, Auto).

Non-Programmable Thermostat: A thermostat that does not have the capability of running *Time Period Programming*.

Programmable Thermostat: A thermostat that has the capability of running *Time Period Programming*.

Reheat: Running the cooling and 2nd stage strip heaters at the same time in order to *dehumidify* the air without significantly cooling down the room temperature.

Temperature Swing: *Same as Deadband.*

Time Period Programming: A program that allows the thermostat to automatically adjust the *heat setpoint* and/or the *cool setpoint* based on the time of the day.

Table of Contents

GET TO KNOW YOUR THERMOSTAT

Get to Know Your Thermostat.....	1
Quick Start	6

INSTALLATION INSTRUCTIONS

Sample Wiring Diagrams.....	12
Test Operation.....	15

USER SETUP

Backlight Operation.....	16
Scrolling Display Options	17
Thermostat Display Options.....	17
Emergency Heat	18
RF Module.....	19
Wi-Fi Module.....	20
System Runtimes.....	23

INSTALLER SETUP

Program Mode Operation.....	26
Programming Fan Operation.....	27
Comfort Recovery Operation	27
Humidification & Dehumidification	28
Dry Contact Operation	29
Remote Sensor Operation.....	30
ADR	31
Locking/Unlocking the Keypad.....	37
Factory Defaults.....	38

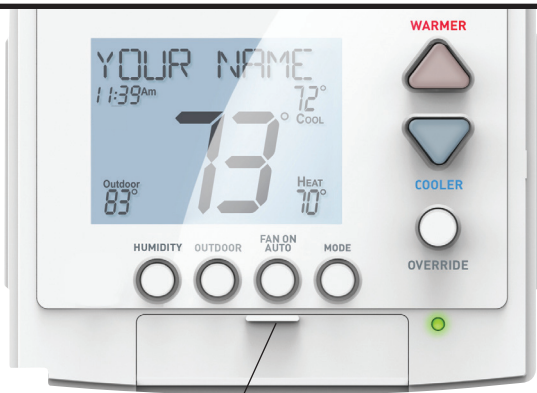
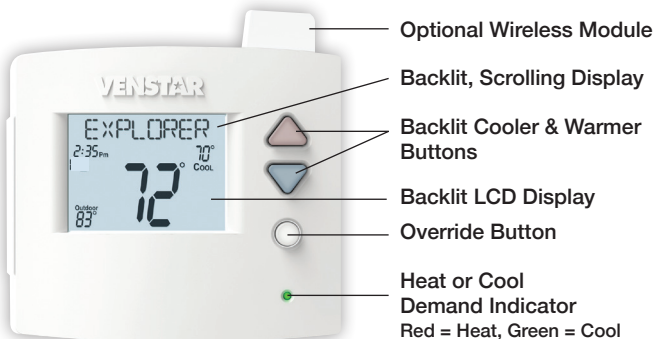
TECHNICIAN SETUP

Sensor Calibration	39
Equipment Testing	39
Advanced Output Testing.....	39
Troubleshooting	40
Advanced Setup Table	41

INDEX

WARRANTY

Get To Know Your Thermostat



Setup Buttons Behind Door

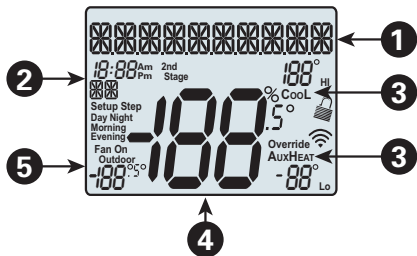
Get To Know Your Thermostat



Setup Buttons

Get To Know Your Thermostat

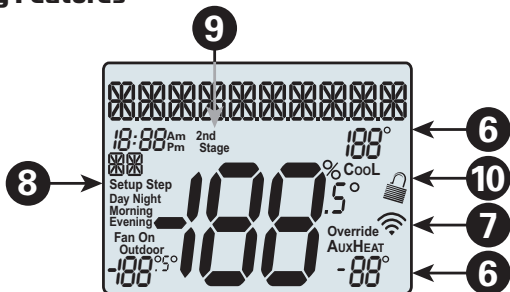
Display Features



- 1 The scrolling display will be used to help you easily navigate the setup screens in the thermostat.
- 2 Clock
Indicates the current time.
- 3 Mode Indicators
Selects the operational mode of the equipment.
HEAT - Indicates the heating mode.
COOL - Indicates the air conditioning mode.
HEAT & COOL - Indicates the system will automatically change-over between heat and cool modes as the temperature varies.
OFF - Indicates heating and cooling is turned off.
- 4 Room Temperature Display
Indicates the current room temperature and displays the outdoor temperature when selected.
- 5 **Outdoor** icon
Indicates the temperature displayed is from the optional outdoor sensor.

Get To Know Your Thermostat

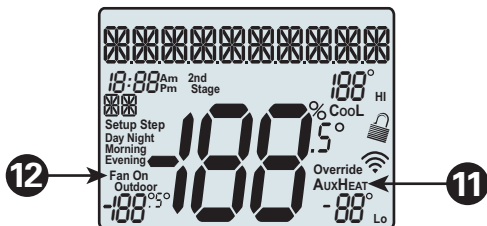
Display Features



- 6** **Desired Set Temperature**
Indicates desired room temperature(s). Also displays the highest and lowest temperatures for the day.
- 7** **Wi-Fi icons**
One dot indicates the thermostat recognizes the wireless module. The “pull” icon indicates the thermostat is currently connected to the Local access point, via the optional Wi-Fi Module.
- 8** **Setup Step icon**
Indicates the step number when the thermostat is in the setup mode.
- 9** **2nd Stage icon**
Indicates what stage of cooling or heating is currently energized.
- 10**  **icon**
Indicates the keypad has been locked.

Get To Know Your Thermostat

Display Features



11 AuxHeat icon

Indicates 2nd stage electric strip heat is being used when the thermostat is programmed for Heat Pump operation. Only the Aux icon will appear during Cool to Dehumidify to indicate Reheat operation.

12 Fan On icon

Indicates constant, continuous fan operation.
When **Fan On** is not lit - indicates the fan will only operate when necessary to heat or to cool.

Quick Start

During Setup and Programming:

Press the WARMER or COOLER buttons to modify the selection.

Press the MODE button to advance and confirm through the setup steps.

Setting the Clock

Not available when wi-fi module is present

Press the SET CLOCK button. Adjust the clock using the WARMER or COOLER buttons. Press the SET CLOCK button to confirm settings.

TIP: To adjust the time by hours press and hold the FAN button while pressing the WARMER or COOLER buttons.



Selecting the Heat or Cool Mode

Select mode by pressing the MODE button.



Heating Only - Only the heating operation will be controlled by the thermostat in this mode.

Cooling Only - Only the cooling operation will be controlled by the thermostat in this mode.

Heating or Cooling (Auto-Changeover) - AUTO will automatically select heat or cool based on room temperature demand.

OFF - OFF indicates both heating and air conditioning systems are turned off.

Quick Start

Selecting your desired temperature

AUTO-CHANGEVER MODE - Pressing the WARMER or COOLER buttons in Auto mode will adjust both the heat and cool setpoints simultaneously. To adjust heat and cool setpoints individually, choose HEAT mode to adjust the heat setpoint and COOL mode to adjust the cool setpoint, then return to AUTO mode.

HEAT OR COOL MODE - Pressing the WARMER or COOLER buttons in Heat or Cool mode will adjust only the heat or cool setpoints individually displayed.

Using the Fan Button

Fan On indicates constant fan operation. Fan On is not allowed when the thermostat is in the OFF mode. Pressing the FAN button toggles this feature. If you don't see "Fan On", the fan is in auto mode and will only turn on during a heat or cool demand.



Using the Override Button

UNOCCUPIED OPERATION -

During a Dry Contact forced unoccupied period, pressing the OVERRIDE button will temporarily force the thermostat into the mode it was in before the Dry Contact forced it into the Unoccupied mode. For example: If the thermostat was in the Auto mode and the Dry Contact forced the thermostat into the Unoccupied mode, then pressing the OVERRIDE button will force the thermostat back into the Auto mode.

The remaining override time will be displayed in the upper left hand corner of the display. The override timer can be set up to a maximum of four (4:00) hours, in increments of 30 minutes. If the timer has been set for the maximum time, the next press of the OVERRIDE button will reset the timer, returning the thermostat to the Unoccupied mode.

To adjust the setpoints for the Unoccupied mode, see page 29.

Current Override Hours (Setup step 10) - This counter keeps track of the number of hours that the thermostat is overridden into Occupied settings. Press FAN to reset.



Quick Start

Viewing the Temperature Sensors

OUTDOOR TEMP - Press the OUTDOOR button to view the current outdoor temperature. If connected to a Skyport account, pressing outdoor button will show the temperatures for your location if you don't have a wired sensor connected.

Press the OUTDOOR button again to view any connected wired sensor (Remote or Supply).

Note: *If no outdoor sensor is connected, and there isn't outdoor temperature via Wi-Fi, then 2 dashes [- -] will appear with the first button press.*

REMOTE/SUPPLY TEMP - Press the **Accessory Status** button to view linked wireless and wired sensors and other accessories. Press the **Accessory Status** button to return to the main screen. Setup step #26 selects the use of the wired temperature sensor.

OUTDOOR



ACCESSORY
STATUS



Viewing the Indoor Humidity Sensor

IMPORTANT: *Allow at least 2 minutes after the thermostat is powered on for the humidity to read correctly.*

Press the HUMIDITY button then the mode button to display the current humidity measured at the thermostat. The room's relative humidity is displayed in the top left corner. The humidification setpoint appears in the larger, center display and can be adjusted using the WARMER or COOLER buttons. Press the MODE button again to view and adjust the dehumidification setpoints. Press the HUMIDITY or MODE button again to confirm settings and return to normal operation.

HUMIDITY



Note: *Due to variations in environmental and equipment conditions, it is not always possible to achieve the desired humidification or dehumidification setpoint.*

Installation Instructions

Remove and Replace the old thermostat

To install the thermostat properly, please follow these step by step instructions. If you are unsure about any of these steps, call a qualified technician for assistance.

- Assemble tools: Flat blade screwdriver, wire cutters and wire strippers.



- Make sure your Heater/Air Conditioner is working properly before beginning installation of the thermostat.
- Carefully unpack the thermostat. Save the screws, any brackets, and instructions.
- Turn off the power to the Heating/Air Conditioning system at the main fuse panel. Most residential systems have a separate breaker for disconnecting power to the furnace.
- Remove the cover of the old thermostat. If it does not come off easily, check for screws.
- Loosen the screws holding the thermostat base or subbase to the wall and lift away.
- If you have a smart phone handy, take a photo of the wiring for future reference.
- Disconnect the wires from the old thermostat. Tape the ends of the wires as you disconnect them, and mark them with the letter of the terminal for easy reconnection to the new thermostat.
- Keep the old thermostat for reference purposes, until your new thermostat is functioning properly.

Installation Instructions

Wire Connections

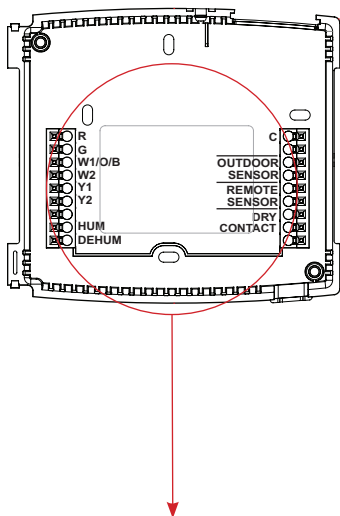
If the terminal designations on your old thermostat do not match those on the new thermostat, **refer to the chart below or the wiring diagrams that follow.**

Wire from the old thermostat terminal marked	Function	Install on the new thermostat connector marked
G or F	Fan	G
Y1,Y	Cooling	Y1
W1,W	Heating	W1/O/B
Rh, R, M, Vr, A	Power	R
C	Common	C
O/B	Rev. Valve	W1/O/B*
W2	2nd Stage Heat	W2
Y2	2nd Stage Cooling	Y2
H, Hum	Humidity	HUM
D, Dehum	Dehumidity	DEHUM
Ck1	Dry Contact Switch	DRY CONTACT
CKGND	Dry Contact Switch	DRY CONTACT

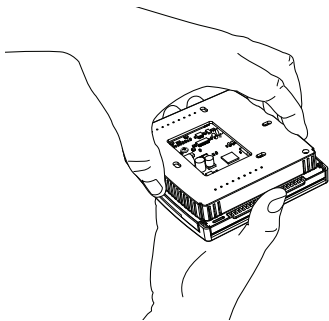
* O/B is used if your system is a Heat Pump.

Installation Instructions

The Explorer Thermostat Backplate



To remove the thermostat backplate:
Gently separate the display from the base by pulling first from one side, then the other until the two pieces unsnap. A small screwdriver may be used, very carefully, to start separating the two pieces.



R	24 VAC return
G	Fan relay
W1/O/B	1st stage heat circuit
W2	2nd stage heat circuit
Y1	1st stage compressor relay
Y2	2nd stage compressor relay
HUM	Humidifier control circuit
DEHUM	Dehumidifier control circuit

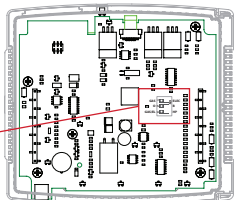
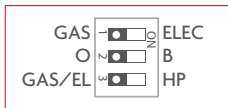
C	24 VAC common
OUTDOOR SENSOR	Outdoor sensor connections
REMOTE SENSOR	Remote sensor connections
DRY CONTACT	Dry Contact connections

IMPORTANT: This thermostat requires both **R (24 VAC Return)** and **C (24 VAC Common)** be connected to the backplate terminals.

Installation Instructions

Check Dip Switch

Ensure which switch is correct for your system. Dip switches are located on the back of the thermostat.



GAS ELEC



OR

GAS ELEC



1. When GAS/EL or HP is set for GAS/EL:

This switch (GAS or ELEC) controls how the thermostat will control the Fan (G) terminal in heating mode. When GAS is chosen, the thermostat will not energize the Fan (G) terminal in heating. When ELEC is chosen the thermostat will energize the fan in heating.

2. When GAS/EL or HP is set for HP:

This switch (GAS or ELEC) defines the Aux Heat type. When GAS is chosen, the auxiliary heat will not be allowed to run during heat pump operation. When ELEC is chosen, up to two stages of auxiliary strip heat will be allowed to run.



OR



For Heat Pump Only

When the GAS/EL or HP dip switch is configured for HP, this dip switch (O or B) must be set to control the appropriate reversing valve. If O is chosen, the W1/O/B terminal will energize in cooling. If B is chosen, the W1/O/B terminal will energize in heating.



OR



GAS/EL HP

GAS/EL HP

This dip switch configures the thermostat to control a conventional gas/electric system or a heat pump. If your system is anything other than a heat pump, leave this switch set for GAS/EL.

Installation Instructions

Sample Wiring Diagrams

Conventional Heating and Cooling Systems

3 Wire, Heat Only

Residential & Commercial 1 Stage Heating with no Fan.

R 24VAC Power
C 24VAC Common
W1/O/B 1st Stage Heat



4 Wire, Cool Only

Residential & Commercial 1 Stage Cooling.

R 24VAC Power
C 24VAC Common
Y1 1st Stage Cool
G Fan



5 Wire, 1 Stage Cooling, 1 Stage Heat

Residential & Commercial 1 Stage Cooling, with 1 stage Gas Heat.

R 24VAC Power
C 24VAC Common
W1/O/B 1st Stage Heat
Y1 1st Stage Cool
G Fan



5 Wire, 1 Stage Cooling, 1 Stage Heat

Residential & Commercial 1 Stage Cooling, with 1 stage Electric Heat.

R 24VAC Power
C 24VAC Common
W1/O/B 1st Stage Heat
Y1 1st Stage Cool
G Fan



Installation Instructions

Sample Wiring Diagrams

Heat Pump Systems

5 Wire, 1 Stage Cooling, 1 Stage Heat

Residential & Commercial Heat Pump with
'O' Reversing Valve

R	24VAC Power
C	24VAC Common
W1/O/B	Reversing Valve
Y1	1st Stage Compressor (Cool or Heat)
G	Fan



6 Wire, 1 Stage Cooling, 2 Stage Heat

Residential & Commercial Heat Pump with
'O' Reversing Valve

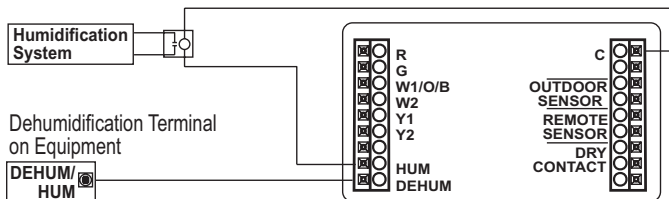
R	24VAC Power
C	24VAC Common
W1/O/B	Reversing Valve
Y1	1st Stage Compressor (Cool or Heat)
W2	Aux Heat
G	Fan



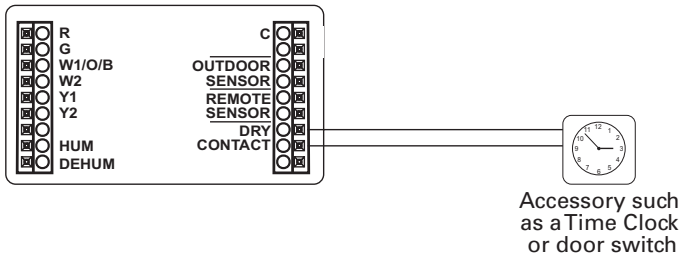
Installation Instructions

Sample Wiring Diagrams

Humidification or Dehumidification



Dry Contact



Installation Instructions: Test Operation

The Explorer thermostat has a diagnostic feature that enables testing of all outputs. This feature is contained in **Technician Setup**.

To enter Technician Setup, press and hold the SETUP button for 10 seconds until all the icons appear. Follow the next steps to view settings and test equipment.

1. Press MODE to view the version numbers of the thermostat.
2. Press MODE again to view the jumper settings and current state of the Dry Contact and Fault terminals.

3. Press MODE again and the scrolling display will read "TURN ON EQUIPMENT?" Press WARMER for Yes or COOLER for No.

If Yes is chosen, press WARMER to turn on heat or COOLER to turn on Cooling. The scrolling display will read "NOTHING ON." Next:

Press WARMER to turn on and cycle up through the heating stages.
Press COOLER to turn the heating stages off. Press MODE to exit.

Press COOLER to turn on and cycle down through the cooling stages.
Press WARMER to turn the cooling stages off. Press MODE to exit.

4. Press MODE until "CALIBRATE SENSORS?" appears on the scrolling display. Press WARMER for Yes or COOLER for No. Press MODE to select which sensor to calibrate. Use WARMER or COOLER to modify your selection.
5. Press MODE until "CONTROL HUM?" appears on the scrolling display. Press WARMER for On or COOLER for Off. Press MODE to continue.
6. Press MODE until "CONTROL DEHUM?" appears on the scrolling display. Press WARMER for On or COOLER for Off. Press MODE to continue.

To exit Technician Setup at any time, press the SETUP button. Technician Setup will automatically exit after 10 minutes if no buttons are pressed.

User Setup - Backlight Operation

How to Change Settings in the Setup Screens

To enter Advanced Setup, press the **SETUP** button, then press **MODE**. Use the **WARMER** or **COOLER** buttons to adjust the value of your selection. Press **MODE** to advance to the next setup step. Press **SETUP** again to leave the setup screens.



Backlight (Setup Steps 2-7)

Backlight (setup step 2)

Off - Backlight turns on with any button press and turns off after 8 seconds.

On - Backlight is on continuously.

Backlight Intensity Level (setup step 3)

The backlight can be adjusted between Off and seven levels of brightness.

Night Dimmer (setup step 4) - Selecting **On** allows for automatic dimming of the display at night.

Night Dimmer Brightness (setup step 5)

Off through seven levels of brightness

Night Dimmer Start Time (setup step 6) - 12:00 am to 12:00 am

Night Dimmer Stop Time (setup step 7) - 12:00 am to 12:00 am

Language (setup step 17)

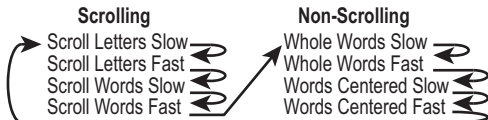
Setup step instructions on the scrolling display can be set for English, Spanish, or French.

Press the SETUP button, then press MODE repeatedly until the Language setup step appears. Use the WARMER or COOLER buttons to make selection. Press MODE to advance to the next step. Press SETUP to leave the setup screens.

User Setup - Scrolling Screen & Display Options

Scrolling Display Method (Setup Step 18)

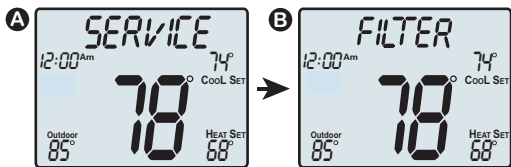
This option allows the user to choose how the scrolling text is displayed. Options are:



Press the SETUP button, then press MODE repeatedly until the Scrolling Method setup step appears. Use the WARMER or COOLER buttons to make selection. Press MODE to advance to the next step. Press SETUP to leave the setup screens.



Example of “Whole Words Centered”:



Emergency Heat

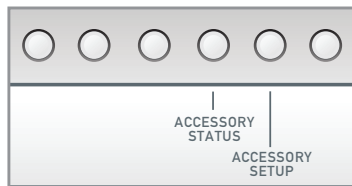
The Emergency Heat function is only available if your thermostat is set to control a Heat Pump.



To initiate the Emergency Heat feature, Press the EMERGENCY button. During Emergency Heat operation the thermostat will turn on the fan and auxiliary stages of heat when there is a demand for heat. The 1st stage of heating and all stages of cooling will be unavailable. To exit Emergency Heat, press the EMERGENCY button.

User Setup

Wireless Module



The **Accessory Status** button allows the user to view the status of wired and wireless accessories. For many of the wireless devices this status includes: Battery Level, Signal Strength & LastTime Updated.

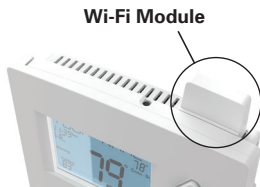
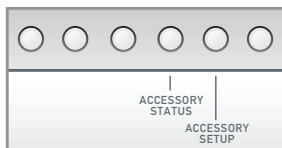
If there is an optional wireless module installed, the **Accessory Setup** button allows the user to link or connect wireless devices to the thermostat, or the thermostat to the network.

Voyager thermostats may use 1 of 2 different types of modules. They are:

1. Wi-Fi Module
2. Z-Wave Plus Module

User Setup

Wi-Fi Module



Please follow the instructions included with the Wi-Fi module to connect to an Access Point or view status. The general instructions are below.

Wi-Fi Module

If the  is present on the display then the thermostat is connected to the Wi-Fi Access Point. If just the “dot” of this icon appears, then just the Wi-Fi module is recognized.

Press the **Accessory Status** button, then press either the Cooler button to view connected Wi-Fi sensors, OR press the Warmer button to view the Wi-Fi status and settings.

Press the Mode button to step through the connected sensors or the Wi-Fi status screens listed below.

- Wi-Fi status (connecting, connected, etc.)
- Signal strength
- Access point name
- IP address
- MAC address
- Skyport status (connecting, connected, etc.)
- Local API status (Enabled, Disabled)
- Module version

• At any time press the **Accessory Status** button to leave the status screens.

Press the Accessory Setup button to enter Wi-Fi or Skyport setup:

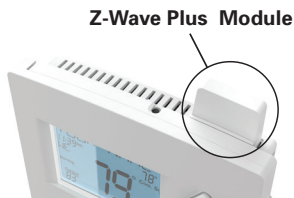
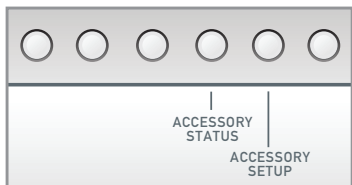
Press the Cooler button to configure Wi-Fi settings.

Press the Warmer button to join this thermostat to a Skyport account. If the thermostat is connected to Wi-Fi and the Internet, a Device ID will appear on the scrolling display of the thermostat. You will enter this code to add this thermostat to your Skyport account via a browser or the Skyport mobile app.

Note: To connect to Skyport Cloud Services, Setup Step #50 must be set to on.

User Setup

Z-Wave Plus Module



Please follow the instructions included with the Z-Wave module to join the Network or view status. The general instructions are below.

Z-Wave Plus Module

Press the **Accessory Status** button to view the status of the thermostat's connection to the Network.

Press the **Accessory Setup** button to enter the Z-Wave Network setup:

Press the Cooler button to join the Z-Wave Network and start the connection process on the Z-Wave controller.

Press the Warmer button to remove this thermostat from the Z-Wave network. When prompted, press the Mode button to remove the thermostat from a connected controller. If the controller is not present, press the Fan button to remove the thermostat forcefully from the network.

Notes regarding Voyager's Z-Wave plus module:

- The Voyager Z-Wave Plus thermostat supports association group 1.
- This group supports up to 5 devices and is used for reporting setpoint changes performed by the user on the thermostat, and for reporting when the thermostat was reset locally by the user.
- The Z-Wave command set allows for the implementation of the thermostat's Home/Away and Vacation settings as described on page 19.
- Products from different manufacturers and product categories can be part of the same network and non-battery powered nodes may act as repeaters, regardless of manufacturers.

User Setup

Note for WiFi and Z WAVE: *In order for 3rd party devices (such as home automation systems) to communicate with your thermostat through its local API, Setup Step #51 must be set to ON.*

If desired both Skyport access and API access may be both set to on, allowing 3rd party device access as well as Skyport access.

User Setup - System Runtimes

These setup steps allow the user to monitor equipment runtimes and program service alerts. Service alerts are displayed in the scrolling marquee.

FAN ON
AUTO



Runtime hours or days
appear in the clock display.



Press and hold FAN to clear
service alert messages from
the scrolling marquee.

Service Filter Runtime (setup steps 8-9, 13-14)

Current Service Filter Runtime Hours (Setup Step 8) - This counter keeps track of the number of hours of fan runtime in the Heating mode, Cooling mode, and in stand alone Fan operation. Press FAN to reset.

Current Service Filter Calendar Days (Setup Step 9) - This counter displays the total number of calendar days that have elapsed since the counter was reset to help the user track Fan runtime. Press FAN to reset.

Set Service Filter Runtime Hours (Setup Step 13) - This timer allows the user to specify the number of hours the fan will run before the "Replace Filter" alert will be displayed. Press COOLER continuously until **0** is displayed to disable this alert.

Set Service Filter Calendar Days (Setup Step 14) - This timer allows the user to specify the number of calendar days that will elapse before the "Replace Filter" alert will be displayed. Press COOLER continuously until **0** is displayed to disable this feature.

Press the **SETUP** button, then press **MODE** repeatedly until the desired setup step appears. Use the **WARMER** or **COOLER** buttons to make selection. Press **MODE** to advance to the next step. Press **SETUP** to leave the setup screens.



User Setup - System Runtimes

*To view, set, or reset System Runtimes, press the **SETUP** button, then press **MODE**. Press **MODE** to advance to the desired setup step. Use the **WARMER** or **COOLER** buttons to adjust the value of your selection. Press **SETUP** again to leave the setup screens.*

UV Lamp Runtime (setup steps 11, 15)

Current UV Lamp Calendar Days (setup step 11) - This counter displays the total number of calendar days that have elapsed since last reset to help the user track UV lamp runtime. Press **FAN** to reset.

Set UV Lamp Calendar Days (setup step 15) - This timer allows the user to specify the number of calendar days the UV Lamp will operate before the "Replace UV Lamp" alert will be displayed. Press **COOLER** continuously until **0** appears to disable this alert.

Humidifier Runtime (setup steps 12, 16)

Current Humidifier Calendar Days (setup step 12) - This counter displays the total number of calendar days that have elapsed since last reset to help the user track the Humidifier run-time. Press **FAN** to reset.

Set Humidifier Calendar Days (setup step 16) - This timer allows the user to specify the number of calendar days the Humidifier will run before the "Service Humidifier" alert will be displayed. Press **COOLER** continuously until **0** appears to disable this alert.

User Setup - Available Modes

Selecting Your Available Modes (setup step 1)

Auto-Changeover - Allows the thermostat to turn on heating or cooling based on room temperature demand. Also allows the manual selection of HEAT only or COOL only and OFF.

Heat and Cool - Allows the thermostat to turn on heating or cooling depending on which one has been manually selected. Auto-Changeover is not available when this is selected.

Heat Only - Allows the thermostat to only turn on HEAT or OFF modes.

Cool Only - Allows the thermostat to only turn on COOL or OFF modes.

Installer Setup

Setpoint Limits (setup step 19) When this feature is at any setting other than no setpoint limits', the heat and cool setpoints can be restricted to preset levels, set in steps 20 and 21.

This feature allows the user to set 3 different levels of security: (0 - 3).

No Setpoint Limits (0) - When this level is selected, no restrictions are activated.

Use Setpoint Limits (1) - When this level is selected, the heat and cool setpoints can be restricted to preset levels, set in setup steps 24 and 25.

Maximum Heat Setpoint (setup step 20) - (35° - 99°).

Minimum Cool Setpoint (setup step 21) - (35° - 99°).

Force Current Mode (2) - When this level is selected, the heat and cool setpoints can be restricted to preset levels, set in setup steps 20 and 21 and the thermostat is locked into the current mode and the FAN button is locked out.

Setpoints Frozen (3) - When this level is selected, the heat and cool setpoints, the current mode and the FAN button are locked.

Cycles Per Hour (setup step 22) The Cycles Per Hour setting may limit the number of times per hour your HVAC unit may energize. For example, at a setting of 6 cycles per hour the HVAC unit will only be allowed to energize once every 10 minutes. The Cycles Per Hour limit may be overridden and reset by pressing the WARMER or COOLER buttons on the thermostat. Settings are No Limit, 2, 3, 4, 5, or 6.

Compressor Minimum Off Minutes (setup step 23) This feature allows the user to set a minimum off time for the compressor. Settings are 5 mins., 3 mins., or 0 mins.

Minimum Heat/Cool Setpoint Difference (setup step 24)

This feature allows the user to set the minimum gap between Heat and Cool setpoints in **AUTO** mode. Select from 0 to 6. If setup step 2 is not set for **AUTO-CHANGEOVER**, this step will not appear.

Number of Heat Stages (setup step 25) This setting assures proper stage callouts on the thermostat display for non-heat pump applications.

Number of Cool Stages (setup step 26) This setting assures proper stage callouts on the thermostat display for non-heat pump applications.

Number of Compressor Stages (setup step 27) *This feature is for heat pump application only.* This feature allows the thermostat to control 1 or 2 compressor stages when configured for heat pump.

Number of Aux Stages (setup step 28) *This feature is for heat pump application only.* This feature allows for proper Aux Heat Staging. (0-2 stages)

Installer Setup

Deadband Settings (setup steps 29)

The Deadband is the number of degrees or minutes that the thermostat waits before it initiates the stages of heating or cooling.

1st Stage Deadband (setup step 29) - Specifies the minimum temperature difference between the room temperature and the desired setpoint before the first stage of heating or cooling is allowed to turn on. (1 - 6 degrees) For example, if the heat setpoint is 68° and the 1st Stage deadband is set to 2 degrees, the room temperature will need to reach **66°** before the heat turns on.

Fan Off Delay in Seconds (setup step 40)

This feature allows the user to increase the cooling or electric strip heating efficiency of the system. The thermostat may be programmed to continue running the fan after a call for cooling or electric strip heating has been satisfied. This delay can be set for 0, 30, 60, 90, or 120 seconds. If set to 0, the fan will not run after a call for cooling or electric strip heating has been satisfied.

Installer Setup

Humidity and Dehumidity (setup steps 33 -38)

Humidity Only With Heat (Setup Step 33) - When this step is set to ON, Humidity will not run without a demand for Heat.

Fan With Humidity Demand (Setup Step 34) - Specifies if the fan should be turned on with a demand for Humidity. (This step will only appear if step 46 is set to OFF.)

Fan with Dehumidify (Setup Step 35) - Specifies if the fan should be turned on with a demand for Dehumidify. (This step will only appear if step 46 is set to OFF.)

Cool To Dehumidify (Setup Step 36) - Specifies if the cooling equipment is allowed to turn on exclusively to lower room humidity. (If set to OFF the following two steps will not appear.)

Max Dehum Overcool (Setup Step 37) - Specifies how many degrees below the Cool setpoint the air conditioning will run to satisfy a Cool to Dehumidify demand. (0° - 20°)

Reheat Operation With Cool To Dehumidify (Setup Step 38) - Specifies if electric strip heat is allowed to turn on during a Cool to Dehumidify demand to help maintain desired room temperature. This step is not available if Electric Heat is not present.

Humidity Output Polarity (setup step 48-49)

Humidity Output Normally Open - means no voltage is sent to the HUM output when there is no demand for humidity.

Humidity Output Normally Closed - means voltage is sent to the HUM output when there is no demand for humidity.

Dehumidify Output Polarity (setup step 48)

Dehumidify Output Normally Open - means no voltage is sent to the DEHUM output when there is no demand to dehumidify.

Dehumidify Output Normally Closed - means voltage is sent to the DEHUM output when there is no demand to dehumidify.

Dehumidify Only with Cooling (setup step 49)

When set to ON, Dehumidify will only turn on with a 1st stage cooling demand. When set to OFF, Dehumidify will turn on at any time that the room humidity exceeds the dehumidification setpoint.

Main Menu Buttons - Information

Dry Contact Operation (setup step 42 - 43)

Dry Contact Polarity (setup step 42)

Open (Normally Open) - The dry contact is open until the connected device closes the circuit.



'Idle'



'Active'

Closed (Normally Closed) - The dry contact is closed until the connected device opens the circuit.



'Idle'



'Active'

Dry Contact Use (setup step 43)

CONDENSATE - If CONDENSATE is selected when the dry contact is active, the thermostat will lockout the compressor terminal(s) and "CONDENSATE PAN OVERFLOW" will appear on the display.

FORCE UNOCCUPIED - If FORCE UNOCCUPIED is selected, the thermostat will use unoccupied mode/ setpoints when the dry contact is active. In that state, the OVERRIDE button is available to return to the prior mode/setpoints for a limited amount of time. This setting is commonly used to emulate the function of a twist timer.

FDD - If FDD is selected when the dry contact is active, "EQUIPMENT FAULT" will appear on the display.

UNOCCUPIED MODE, UNOCCUPIED COOL SETPOINT, UNOCCUPIED HEAT SETPOINT (setup steps 44-46) - When prior step set for FORCE UNOCCUPIED, these three steps appear. Specify the desired mode and setpoints to be used when the dry contact is active.

Wired Sensor Type (setup step 30) - Specifies the use of the connected, wired sensor. The choices are: Remote, Supply, Outdoor. Only the remote option allows control of the sensor.

Installer Setup

Control To Temp Source (setup step 31)

This feature allows the use to specify which temperature sensor source(s) to be used to measure room temperature for control

Thermostat: Uses the internal thermostat sensor only.

Wired Remote: Uses external temperature sensor wired to the REMOTE SENSOR contacts.

Wireless Remote: Uses one wireless remote temperature sensor. Choose which linked sensor to use in the subsequent step.

Average of Wireless Remotes: Averages the temperatures of all linked wireless remote sensors.

Average Thermostat and Wired Remote: Averages the temperatures of the wired remote sensor and the thermostat.

Average All Sensors: Averages the temperatures of the wired remote, any linked wireless remotes and the thermostat.

Wireless Remote To Use (setup step 32)

Specifies the use of the connected, wired sensor. The choices are: Remote or Supply. The remote option allows control to the sensor, the supply does not.

Fahrenheit or Celsius (setup step 41)

This feature allows the thermostat to display temperature in Fahrenheit or Celsius.

Press Fan To Clear All Messages (setup step 60)

This feature allows the user to clear all current error messages from the display.

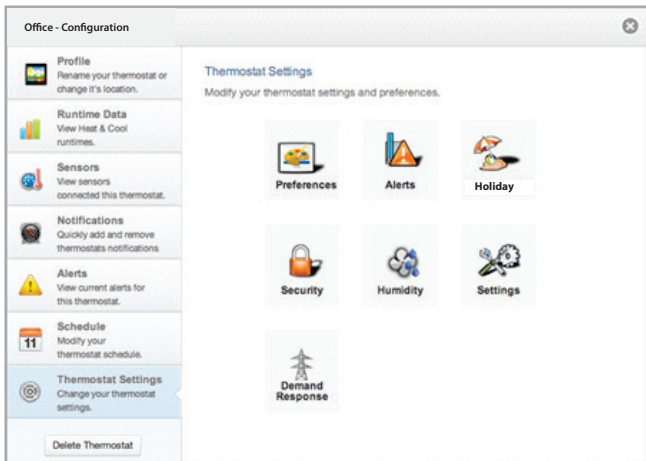
Installer Setup - Automated Demand Response

Overview

Voyager thermostats support the handling of specific signals from the utility provider. The utility generated signals carry pricing information and/or setback actions that alter the comfort settings of the thermostat in order to reduce energy usage on demand. This is known as Automated Demand Response or ADR for short. You must register to participate in a utility sponsored program, if offered by your local utility, to take advantage of this feature.

Skyport Cloud Services

From the web application the user will select Thermostat Settings from the left column. Then the Demand Response button is selected.



Installer Setup - Automated Demand Response

The Demand Response configuration page, shown below, is where the thermostat is configured to respond to the energy provider's signals. It also sets operational parameters for the thermostat.

The left column of the ADR configuration page allows or prevents access by the utility. Here communication with the utility and your thermostat may be turned On or Off.

The selected utility will provide a Program ID.

The Account ID would normally be the occupants utility account number.

Office - Configuration

Profile
Rename your thermostat or change its location.

Runtime Data
View Heat & Cool runtimes.

Sensors
View sensors connected this thermostat.

Notifications
Quickly add and remove thermostats notifications

Alerts
View current alerts for this thermostat.

Schedule
Modify your thermostat schedule.

Thermostat Settings
Change your thermostat settings.

Delete Thermostat

Automated Demand Response

Configuration Overview



Demand Response

☒ ON ☐ OFF

[What is Demand Response?](#)

Select Utility
LADWP

Program Id.
test-program

Account Id.
1234

Cancel

Min & Max Settings

Event Max Cool Setpoint: 85°F

Event Min Heat Setpoint: 65°F

Static Settings

Static Cool Setpoint: 78°F

Static Heat Setpoint: 65°F

Offset Settings

Cool Setpoint Offset: +2

Heat Setpoint Offset: -2

Price Settings

Price Trigger: 45

Price Dependent Action: Observe Setpoint Offsets

Save

Installer Setup - Automated Demand Response

The right column of the ADR configuration page is where the occupant adjusts the operational parameters for ADR. The utility may send up to 3 types of ADR signals to Skyport. These are:

1) Pricing for the cost of energy, **2) An Offset** to the occupants' comfort setpoints, and **3) a signal** to enforce discrete or Static setpoints.

The Maximum Cooling Setpoint and Minimum Heating Setpoints for ADR events are adjusted here.

The Static Settings are applied when the utility sends a signal to allow the occupant to enforce their own discrete temperature settings during an ADR event.

The Offset Settings allow the utility to modify the Cool or Heat setpoints by the value set here during an ADR event.

A Price Trigger setting allows the occupant to set the maximum cost of energy threshold. When this threshold is exceeded the Price dependent action is enforced. This Price Trigger and Dependent action is enforced independent of an ADR event, as long as the utility sends 'real-time' pricing.

Installer Setup - Automated Demand Response

Selecting the Overview tab of the ADR page will cause a summary of ADR events to be displayed.

The screenshot shows a software interface with a sidebar on the left titled "Office - Configuration" and a main content area on the right. The sidebar contains several menu items: Profile, Runtime Data, Sensors, Notifications, Alerts, Schedule, and Thermostat Settings. The "Thermostat Settings" item is highlighted. Below the sidebar is a "Delete Thermostat" button. The main content area displays a window titled "Automated Demand Response" with two tabs: "Configuration" and "Overview". The "Overview" tab is active, showing a table of ADR events. The table has five columns: Event Id, Event Type, Status, Start Time, and Stop Time. There are two rows of data. Below the table are "Cancel" and "Save" buttons. At the bottom of the main content area, there are icons for "Security", "Humidity", "Settings", and "Demand Response".

Office - Configuration

- Profile**
Rename your thermostat or change it's location.
- Runtime Data**
View Heat & Cool runtimes.
- Sensors**
View sensors connected this thermostat.
- Notifications**
Quickly add and remove thermostats notifications
- Alerts**
View current alerts for this thermostat.
- Schedule**
Modify your thermostat schedule.
- Thermostat Settings**
Change your thermostat settings.

Delete Thermostat

Automated Demand Response

Configuration Overview

Event Id	Event Type	Status	Start Time	Stop Time
1233407844	Price	Success	08/10/2014 1:00pm	08/10/2014 2:00pm
1233407844	Price	Success	08/15/2014 4:00pm	08/15/2014 5:00pm

Cancel Save ✓

Security Humidity Settings Demand Response

Installer Setup - Automated Demand Response

ADR (setup step 52)

Controls whether you want the thermostat to possibly respond to signals from the utility provider. Select ON to allow this and to have steps 77-83 appear.

ADR Action (setup step 53)

Allows the user to determine what action is taken when an ADR event is received.

Observe Setpoint Offsets – will offset the heat and cool setpoints by the amounts specified in setup steps 58 and 59

Observe Static Setpoints – will set the heat and cool setpoints to the values specified in setup steps 56 and 57

Event Max Cool Setpoint (setup step 54)

Event Min Heat Setpoint (setup step 55)

Specifies the range of allowable setpoint adjustments to be enforced when any ADR signal has been received from the utility. Since you might be paying more for energy while an event is active, you can impose tighter limits on setpoint ranges that are only enforced during the event.

Static Cool Setpoint (setup step 56)

Static Heat Setpoint (setup step 57)

Specifies the setpoints that will come into use during an event when the ADR ACTION is set to OBSERVE STATIC SETPOINTS.

Installer Setup - Automated Demand Response

Cool Setpoint Offset (setup step 58)

Heat Setpoint Offset (setup step 59)

Specifies how much the current setpoints in effect prior to an event will be altered during an event when the ADR ACTION is set to OBSERVE SETPOINT OFFSETS. The heat setpoint can be automatically lowered by 1 to 10 degrees while the cool setpoint can be automatically raised by 1 to 10 degrees

DISPLAY INDICATIONS WHEN AN ADR EVENT IS HAPPENING

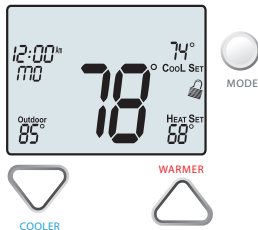
After setting your desired values for use during an ADR event, the scrolling display will give a little information when an event is pending or active. For instance, when an ADR event has been sent to your thermostat, you might see ADR STARTS 8/14 at 2:00pm to notify you of a pending event. Once active, you might see ADR STOPS 8/14 at 6:00pm. When an event is active, you can press any of COOLER, WARMER or MODE buttons, followed by the WARMER to opt out of the event.

Installer Setup

Locking/Unlocking the Keypad

To prevent unauthorized use of the thermostat, the front panel buttons may be disabled. To disable, or 'lock' the keypad, press and hold the MODE button. While holding the MODE button, press the WARMER and COOLER buttons together. The  icon will appear on the display, then release the buttons.

Press all three buttons in the order outlined above for keypad lockout



To **unlock** the keypad, press and hold the MODE button. While holding the MODE button, press the WARMER and COOLER buttons together. The  icon will disappear from the display, then release the buttons.

Installer Setup

Resetting the Thermostat to the Factory Default Settings

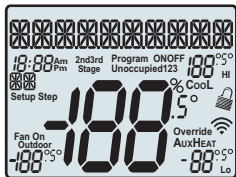
(for default values see page 41-42)

If, for any reason, you desire to return all the stored settings back to the factory default settings, follow the instructions below.

WARNING: This will reset all Time Period and Advanced Programming to the default settings. Any information entered prior to this reset may be permanently lost. Additionally, if a Z-Wave module is installed, resetting the thermostat will not reset the Z-Wave module. To reset the Z-Wave module, please follow the instructions on Page 19.

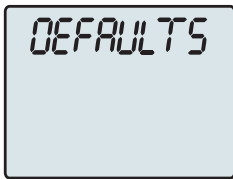
- 1 Press and hold SETUP for 10 seconds. All icons will appear on the display.

Keep pressing the SETUP button until you see this screen.



- 2 After all the icons appear, release SETUP. Press and hold FAN for 5 seconds. DEFAULTS will appear on the display.

Keep pressing the FAN button until you see this screen.



- 3 After DEFAULTS appears, release FAN. Press SETUP to return to normal operation.



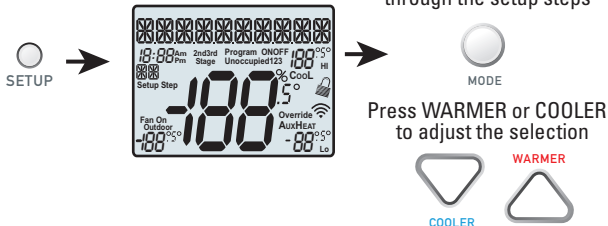
Technician Setup

To enter Technician Setup, press and hold the SETUP button for 10 seconds. After all the icons appear, press MODE. The version number of the thermostat will appear in the scrolling text. Press MODE to advance to the next step. Use the WARMER or COOLER buttons to adjust the value of your selection. To leave Technician Setup, press SETUP.

Hold for 10 seconds

All icons appear

Press MODE to advance through the setup steps



Technician Setup is for diagnostic and testing purposes and is intended for use by a qualified technician. See page 14 for more detailed instructions.

Technician Setup contains the following options:

- View the version number of the thermostat.
- View the Dip Switch equipment type settings.
- View the state of the Dry Contact.
- Turn on equipment outputs for testing.
- Calibrate thermostat, remote, and humidity sensors.
- Control HUM output (On or Off)
- Control DEHUM output (On or Off)

Troubleshooting

- **SYMPTOM:** The air conditioning does not attempt to turn on.
CAUSE: The compressor timer lockout may prevent the air conditioner from turning on for a period of time.
REMEDY: Consult the Owner's Manual in the Installer Setup section to defeat the Cycles Per Hour (*page 22*).
- **SYMPTOM:** The display is blank.
CAUSE: Lack of proper power.
REMEDY: Make sure the power is on to the furnace and that you have 24vac between **R & C**.
- **SYMPTOM:** The air conditioning does not attempt to turn on.
CAUSE: The cooling setpoint is set too high.
REMEDY: Lower the cooling setpoint or lower the cooling set-point limit.
See Setpoint Limits (page 22).
- **SYMPTOM:** The heating does not attempt to turn on.
CAUSE: The heating setpoint is set too low.
REMEDY: Raise the heating setpoint or raise the heating set-point limit.
See Setpoint Limits (page 22).
- **SYMPTOM:** When controlling a residential heat pump, and asking for cooling, the heat comes on.
CAUSE: The thermostat reversing valve jumper is set for **"B"**.
REMEDY: Set the reversing valve jumper for **"O"**.
- **SYMPTOM:** When calling for cooling, both the heat and cool come on.
CAUSE: The thermostat equipment jumper is configured for **"HP"** and the HVAC unit is a Gas/Electric.
REMEDY: Set the equipment jumper for **"Gas"**.
- **SYMPTOM:** When the Program button is pressed, the display reads **"DISABLED"**.
CAUSE: Program mode is set to **"NON PROGRAM"**.
REMEDY: Set Program Mode (Setup 1) to 1, **5/2**, or **7 Day**.
See Selecting Your Program Mode (page 21).

Advanced Setup Table

Df = Factory Default Setting

Step#	Description	Pg#	Range	Df
1	Available Modes	25	Heat/Cool/Auto/Off, Heat/Cool/Off, Heat/Off, Cool/Off	Heat/ Cool/ Auto/Off
2	Backlight	17	On, Off	Off
3	Backlight Level	17	Off thru 7 levels of brightness	Level 5
4	Night Dimmer	17	On/Off	Off
5	Night Dimmer Brightness	17	Off thru 7 levels of brightness	2 (20%)
6	Night Dimmer StartTime	17	12A-12A	8:00P
7	Night Dimmer StopTime	17	12A-12A	6:00A
8	Current Service Filter Runtime Hours	23	0-1999 Hours	0
9	Current Service Filter Calendar Days	23	0-720 Days	0
10	Current Override Hours	7	0-1999 Hours	0
11	Current UV Lamp Calendar Days	24	0-720 Days	0
12	Current Humidifier Calendar Days	24	0-720 Days	0
13	Set Service Filter Runtime Hours	23	0-1950 hours	0
14	Set Service Filter Calendar Days	23	0-720 Days	0
15	Set UV Lamp Calendar Days	24	0-720 Days	0
16	Set Humidifier Calendar Days	24	0-720 Days	0
17	Language	17	English, Espanol, Francais	English
18	Scrolling Method	18	"L-R Slow, L-R Fast, Word L-R Slow, Word L-R Fast, Whole Word L Slow, Whole Word R Slow, Whole Word Ctr. Fast, Whole Word Ctr. Slow"	"Whole Words Center Fast"
19	Setpoint Limits	26	0 - 3	0
20	Max Heat Setpoint	26	35 - 99 Degrees	74
21	Min Cool Setpoint	26	35 - 99 Degrees	70
22	Cycles Per Hour	26	No Limit, 2, 3, 4, 5, 6	6
23	Compressor Minimum Off Minutes	26	0, 3, 5 Minutes	5
24	Min. Heat/Cool Setpoint Difference	26	0 - 6 Degrees	2
25	Number of Heat Stages	26	0 - 2	2
26	Number of Cool Stages	26	0 - 2	1
27	Number Of Compressor Stages	26	1, 2	1
28	Number of Aux Stages	26	0, 1, 2	0
29	1st Stage Deadband	27	1 - 6 Degrees	2
30	Wired SensorType	29	Remote, Supply	Remote

cont. next page

Advanced Setup Table

Df = Factory Default Setting

Step#	Description	Pg#	Range	Df
31	Control to Temp Source	30	Thermostat, Wired Remote*, Wireless Remote, Average of Wireless Remotes, Average Thermostat and Wired Remote*, Average All Sensors. *Option only if prior step = "Remote"	Thermostat
32	Wireless Remote to Use	30	list of wifi sensors currently linked to thermostat. *This step only appears if prior step = "Wireless Remote"	first linked sensor in list
33	Humidity Only With Heat	28	On, Off	Off
34	Fan With Humidity Demand	28	Fan On, Fan Off	Fan Off
35	Fan With Dehumidify Demand	28	Fan On, Fan Off	Fan Off
36	Cool To Dehumidify	28	On, Off	Off
37	Maximum Occ Dehum Overcool	28	0 - 20 Degrees	2
38	Maximum Unocc Dehum Overcool	28	0 - 20 Degrees	2
39	Reheat Operation W/Cool To Dehum.	31	On, Off	Off
40	Fan Off Delay	27	0 - 120 Seconds	0
41	F/C	30	Fahrenheit (F), Celsius (C)	F
42	Dry Contact Polarity	29	Open, Closed	Open
43	Dry Contact Use	29	Condensate Pan, Unoccupied, FDD	Unoccupied
44	Unoccupied Mode	29	Heat, Cool, Auto, Off	Off
45	Unoccupied Cool Setpoint	29	35 - 99 Degrees	85
46	Unoccupied Heat Setpoint	29	35 - 99 Degrees	55
47	Humidity Polarity	28	Open, Closed	Open
48	Dehumidify Polarity	28	Open, Closed	Open
49	Dehumidify only with Cooling	28	On, Off	On
50	Skyport	21	On, Off	On
51	Local API	23	On, Off	Off
52	ADR	35	On, Off	On
53	ADR Action	35	Observe Setpoint Offset, Observe Static Setpoints	Observe set-point offsets
54	Event Max Cool Setpoint	35	65 - 90	90
55	Event Min Heat Setpoint	35	50 - 85	50
56	Static Cool Setpoint	35	65 - 85	82
57	Static Heat Setpoint	35	65 - 85	60
58	Cool Setpoint Offset	36	1 to 10	4
59	Heat Setpoint Offset	36	-1 to -10	-4
60	Press Fan To Clear All Messages	30		

Index

A

Alerts

see Runtime

Auto

adjust temperature, 6
changeover, 7
fan, 5, 7
mode, 3, 7

Aux Icon, 5

AuxHeat icon, 5

Average

thermostat sensor, 30

B

b reversing valve, 10, 11, 12, 14

Backplate, 11

Buttons

accessory, 2, 19-22
cooler (down) 1, 7
emergency heat, 2, 18
front panel, 1
humidity, 1, 8
mode, 1, 6
outdoor, 1, 8
override 1, 7
set clock, 1, 6
setup, 1, 17
up (warmer), 1, 7

C

C, 27

Calibrate Sensors, 16

Celsius, 30

Clock

display, 2
setting, 5

Cool

1st stage deadband,
see Deadband
dehum, 25
minutes of runtime, 23
droop, *see Deadband*
electric/heat pump, 10
icon, 2
indicator, 1
mode, 2, 5
overcool, *see Overcool*
runtime, *see Runtime*
setpoint, 3, 6, 20
to dehumidify, *see*
Dehumidify

Condensate Drain Pan, 29

Cycles Per Hour, 26

D

Deadband

balance point, 27
1st stage, 27

Dehumidify

Aux icon, 2
cool to, 28
dehum settings, 28
setpoint, 28

Delay

fan-off, *see* Fan

Differential

heat and cool, 22

Dip Switch, 12

ELEC, 10
electric heat, 10
gas electric, 10
heat pump, 10
reversing valve, 10
viewing, 33

Disabled Keypad

see Keypad Lockout

Drain Pan Overflow

Alarm, *see Dry*
Contact

Dry Contact

operation, 29
polarity, 29
service pan, 29

Index

E

EH, 32

Emergency Heat, 18

Electric Heating

AuxHeat icon, 3
jumper setting, 10
reheat, 25

F

Factory Defaults

caution, i
settings, 34, 35
resetting, 32

Fahrenheit, 30

Fan

mode, 24
on during heat, *see*
 Electric Heat
on icon, 5
runtime, 18
2nd stage heat, *see*
 Emergency Heat
speed, *see* Dehumidify
with humidity, 28

FDD, 29

G

Gas/Electric Furnace

dip switch, 12

Green Indicator, 1

H

Heat

1st stage deadband,
 see Deadband
emergency heat, 18
min. of runtime, 23
AuxHeat icon, 3
deadband, *see*
 Deadband
droop, *see Deadband*
electric/heat pump, 12
icon, 4
indicator, 1
mode, 3
runtime, *see Runtime*
setpoint, 6, 20

Heat Pump

AuxHeat, 12, 13, 26
Dip Switch Setting, 12
emergency heat, 17
multi-stage, 12, 22

Humidity

service, 19
setpoint, 25
with Fan, 25

K

Keypad Lockout, 31

L

LCD, 1

Language, 17

Locked Indication,

see Keypad Lockout

Lo

icon, 4
temperature, 6

Index

M

Maximum Outdoor Temperature, *see Hi*

Minimum Outdoor Temperature, *see Lo*

Mode, 2, 5, 21

Multi-stage Operation, 23

N

Non-Programmable Thermostat, 21

Normally Open/Closed,
dry contact, 29
dehum terminal, 28

O

O Reversing Valve
see Reversing Valve

Occupied 3, 20

Off Mode, 2, 5

Outdoor
button, *see Buttons*
icon, 1
sensor, 6
viewing temperature, 6

Overcool, 28

Override, 7

P

Pan, Service
see Dry Contact

Polarity, *see Dry Contact*

R

Reheat
during cool to
dehumidify, 28
electric heating, 28
function, 28

Remote Sensor
averaging with
Thermostat, 30
calibrate, 32
control to, 27
outdoor temperature,
see Outdoor
read to, 6
viewing, 6

Reset
Alert Messages, 18
thermostat settings,
see Factory Defaults
runtime
fan/filter, 18
humidify, 19
UV light, 19

Reversing Valve, 10, 14

Runtime
resetting, *see Reset*
setting,
humidifier, 19
service filter, 23
UV light, 24
viewing,
cool, 19
heat, 19
humidification, 19
UV lamp, 19

Index

S

2nd stage turn off temperature, 23

Sensor

outdoor, *see Outdoor*
remote, *see Remote*
thermostat, *see Thermostat*

Service

filter icon, *see Reset*
humidify icon, *see Reset*
pan icon, *see Dry*
Contact
UV light, *see Reset*

Set Clock, *see Clock*

Setpoint

cool, *see Cool*
dehumidification, 25
heat, *see Heat*
humidification, 25

Simplest Operation, 5

Stages

number of heat stages, 26
number of cool stages, 26

T

Technician Setup, 14, 33

Thermostat Sensor

averaging, 30
calibrate, 16, 33

Time, *see Clock*

Time Delay,

compressor lockout, 36
cycles per hour, 22, 26, 34, 36
1st to 2nd stage, 23

U

Unoccupied, 7, 29, 35

UV Lamp

runtime, *see Runtime*
setting, *see Runtime*

W

Warranty, 41

Wi-Fi, 20

Z

Z-Wave, 21

Warranty

One-Year Warranty - This Product is warranted to be free from defects in material and workmanship. If it appears within one year from the date of original installation, whether or not actual use begins on that date, that the product does not meet this warranty, a new or remanufactured part, at the manufacturer's sole option to replace any defective part, will be provided without charge for the part itself provided the defective part is returned to the distributor through a qualified servicing dealer.

THIS WARRANTY DOES NOT INCLUDE LABOR OR OTHER COSTS incurred for diagnosing, repairing, removing, installing, shipping, servicing or handling of either defective parts or replacement parts. Such costs may be covered by a separate warranty provided by the installer.

THIS WARRANTY APPLIES ONLY TO PRODUCTS IN THEIR ORIGINAL INSTALLATION LOCATION AND BECOMES VOID UPON REINSTALLATION.

LIMITATIONS OF WARRANTIES – ALL IMPLIED WARRANTIES (INCLUDING IMPLIED WARRANTIES OF FITNESS FOR A PARTICULAR PURPOSE AND MERCHANTABILITY) ARE HEREBY LIMITED IN DURATION TO THE PERIOD FOR WHICH THE LIMITED WARRANTY IS GIVEN. SOME STATES DO NOT ALLOW LIMITATIONS ON HOW LONG AN IMPLIED WARRANTY LASTS, SO THE ABOVE MAY NOT APPLY TO YOU. THE EXPRESSED WARRANTIES MADE IN THIS WARRANTY ARE EXCLUSIVE AND MAY NOT BE ALTERED, ENLARGED, OR CHANGED BY ANY DISTRIBUTOR, DEALER, OR OTHER PERSON WHATSOEVER.

ALL WORK UNDER THE TERMS OF THIS WARRANTY SHALL BE PERFORMED DURING NORMAL WORKING HOURS. ALL REPLACEMENT PARTS, WHETHER NEW OR REMANUFACTURED, ASSUME AS THEIR WARRANTY PERIOD ONLY THE REMAINING TIME PERIOD OF THIS WARRANTY.

THE MANUFACTURER WILL NOT BE RESPONSIBLE FOR:

1. Normal maintenance as outlined in the installation and servicing instructions or owner's manual, including filter cleaning and/or replacement and lubrication.
2. Damage or repairs required as a consequence of faulty installation, misapplication, abuse, improper servicing, unauthorized alteration or improper operation.
3. Failure to start due to voltage conditions, blown fuses, open circuit breakers or other damages due to the inadequacy or interruption of electrical service.
4. Damage as a result of floods, winds, fires, lightning, accidents, corrosive environments or other conditions beyond the control of the Manufacturer.
5. Parts not supplied or designated by the Manufacturer, or damages resulting from their use.
6. Manufacturer products installed outside the continental U.S.A., Alaska, Hawaii, and Canada.
7. Electricity or fuel costs or increases in electricity or fuel costs for any reason whatsoever including additional or unusual use of supplemental electric heat.
8. ANY SPECIAL INDIRECT OR CONSEQUENTIAL PROPERTY OR COMMERCIAL DAMAGE OF ANY NATURE WHATSOEVER. Some states do not allow the exclusion of incidental or consequential damages, so the above may not apply to you.

This warranty gives you specific legal rights and you may also have other rights which may vary from state to state.

Programming Worksheet - see page 20

DAY	PERIOD	START TIME	COOL	HEAT
MONDAY	Unoccupied			
	Occupied 1			
	Occupied 2			
	Occupied 3			
TUESDAY	Unoccupied			
	Occupied 1			
	Occupied 2			
	Occupied 3			
<i>Copy Mon to Tues</i> ☐ No ☐ Yes				
WEDNESDAY	Unoccupied			
	Occupied 1			
	Occupied 2			
	Occupied 3			
<i>Copy Tues to Wed</i> ☐ No ☐ Yes				
THURSDAY	Unoccupied			
	Occupied 1			
	Occupied 2			
	Occupied 3			
<i>Copy Wed to Thurs</i> ☐ No ☐ Yes				
FRIDAY	Unoccupied			
	Occupied 1			
	Occupied 2			
	Occupied 3			
<i>Copy Thurs to Fri</i> ☐ No ☐ Yes				
SATURDAY	Unoccupied			
	Occupied 1			
	Occupied 2			
	Occupied 3			
<i>Copy Fri to Sat</i> ☐ No ☐ Yes				
SUNDAY	Unoccupied			
	Occupied 1			
	Occupied 2			
	Occupied 3			
<i>Copy Sat to Sun</i> ☐ No ☐ Yes				

Patent Pending



Printed on recycled paper.

P/N 88-1014 Rev. 3 10/17

VENSTAR®